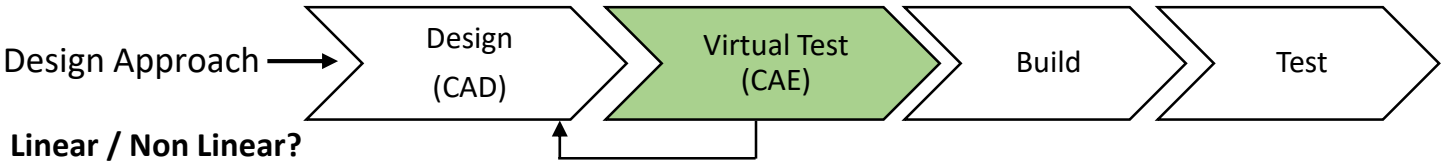
CAE Report 2026

College Name – D. Y. Patil International University, Akurdi, Pune

Team Name & ID – Team Genesis 16 Motorsports (262019)

CAE Fundamentals and Knowledge

Computer-Aided Engineering (CAE) uses structural, thermal, dynamic, and fatigue simulations to predict component behaviour before prototyping. Through FEA principles—material modelling, boundary conditions, and solver interpretation—it enables optimized design and reduced prototyping. Strong fundamentals in meshing, load application, result validation, and safety factors ensure reliability and safety under all operating conditions.



Linear / Non Linear?

No.	Material	Material Properties	Geometry	Contact Influence	Overall Classification
1	**Al Series 6065**	Linear elastic below yield	Linear unless large deformation	Contact does not affect material linearity	**Linear**
2	**AL 7075**	Linear elastic below yield	Linear unless large deformation	Contact does not affect material linearity	**Linear**
3	**EN 36**	Linear elastic	Generally linear	Linear	**Linear**
4	**EN 19**	Linear elastic	Linear	Linear	**Linear**
5	**Mild Steel (MS)**	Linear elastic	Linear	Linear	**Linear**
6	**SS 410**	Linear elastic	Linear	Linear	**Linear**
7	**AISI 4130**	Linear elastic	Linear unless slender geometry buckles	Linear	**Linear**
8	**Carbon FIber**	Non Linear	Can show large strain → nonlinear	Contact (friction) increases nonlinearity	**Non-Linear**

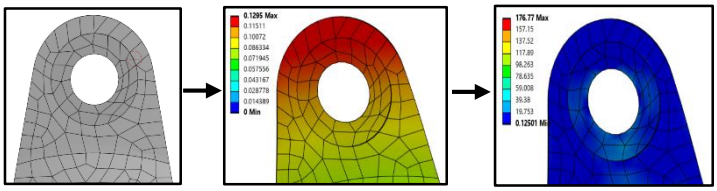
Parameters considered for Rollcage Analysis

Parameters	Values	Parameters	Values
Skewness	60	Warpage	5
Aspect Ratio	5	Jacobian	0.60
Non-Orthogonality	45		

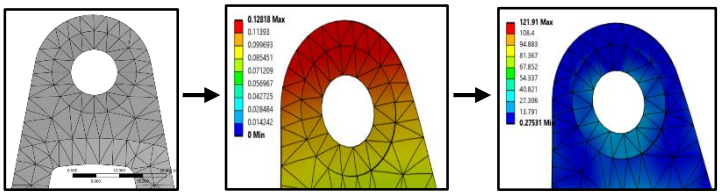
Mesh Size v/s Results (FEA on Wheel Hub)

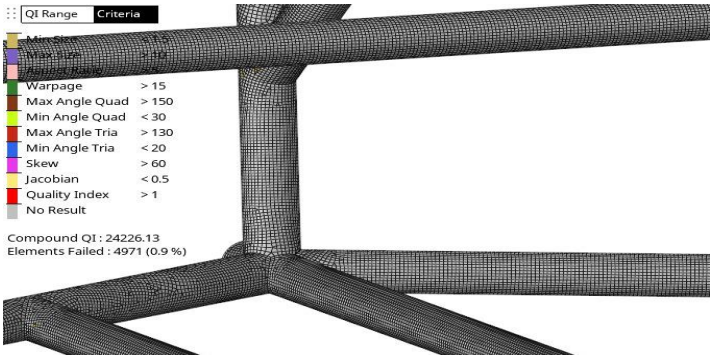
Element size	Nodes	Deformation	Stress
2 mm	160283	0.129 mm	146.93 Mpa
3 mm	54568	0.129 mm	146.24 Mpa
5 mm	16634	0.128 mm	131.31 Mpa
7 mm	8798	0.126 mm	129.47 Mpa

Mesh Type – Hex Dominant



Mesh Type – Tetrahedron

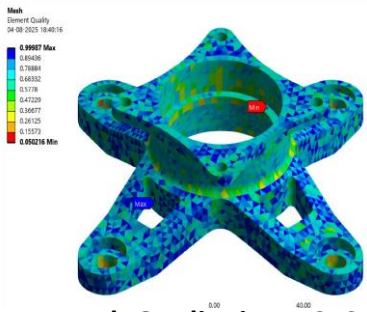




Mesh Quality in Hyperworks

Hex Dominant meshing was chosen as it offered higher - quality elements and Improved convergence.

Failure Theories / Criteria	
Ductile Material	Von Misses Theory
Brittle Material	Principal Stresses
Fatigue Failure	Goodman / Soderberg Criteria
Composite Material	Tsai – Wu Criterion
Reversal Loading	Modified Soderberg Criteria



Mesh Quality in ANSYS

Geometry Cleanup - Geometry cleaned, removing unwanted small fillets, pit holes, and repairing surface defects for better Mesh Quality and results

ROLL-CAGE ANALYSIS

Methodology

1. Roll cage line-diagram CAD model was prepared.
2. Geometry was imported into HyperWorks.
3. Node connections were created and tubing (piping) was generated.
4. Meshing of the roll cage members was performed.
5. Boundary conditions and loading were applied.
6. Structural analysis was carried out to obtain stresses, deformation, and FOS.

Processing Parametres

Pre - Processing
Altair Hyperworks

Solver
Optistruct – Static
Radioss – Dynamic

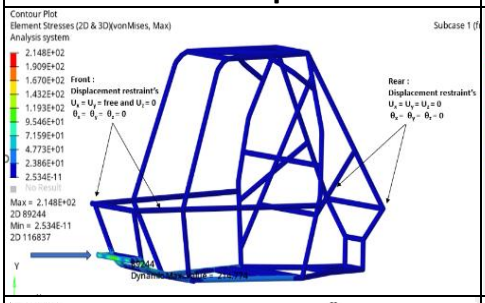
Post - Processing
Altair Hyperview

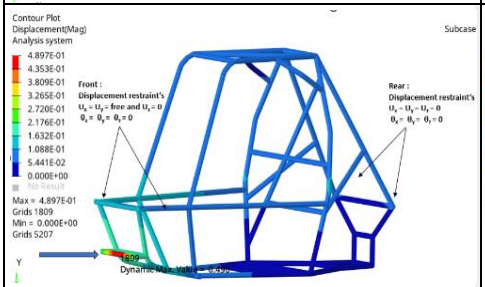
Material Used – Annealed AISI 4130

Density	7850 kg/m
Young's Modulus	205 GPa
YTS	435 Mpa
UTS	560 Mpa

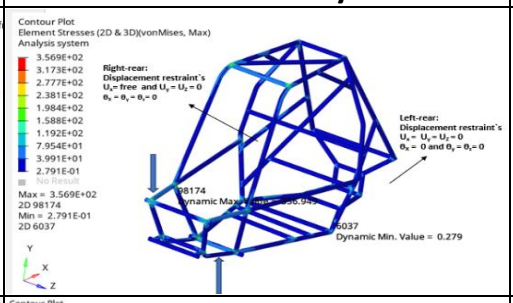
Front Impact $= \frac{m\,\Delta v}{\Delta t}$ = 12.24 KN	Roll Over **(F)** = $\frac{W}{\delta} = \frac{6572.82}{1.66}$ = 3.93 KN	Torsional=T = $\frac{F}{[0.5 \times (C_{tx}\, x\, V)]}$ = 8.3 KN

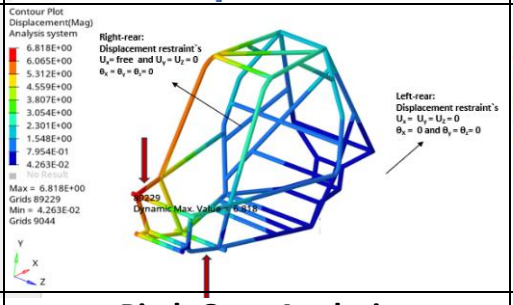
Front Impact



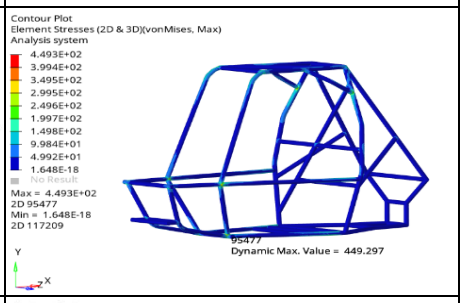


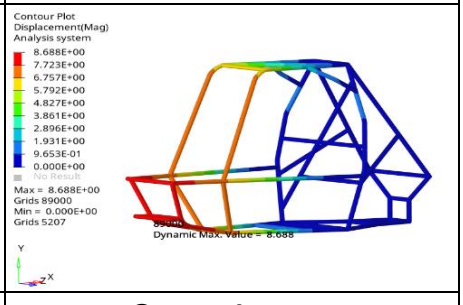
Torsional Analysis



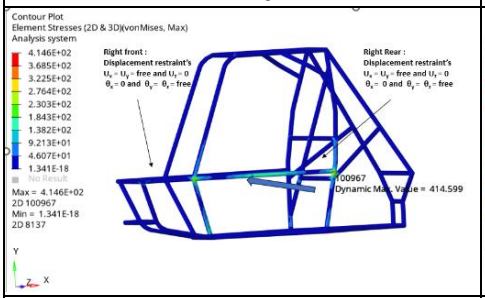


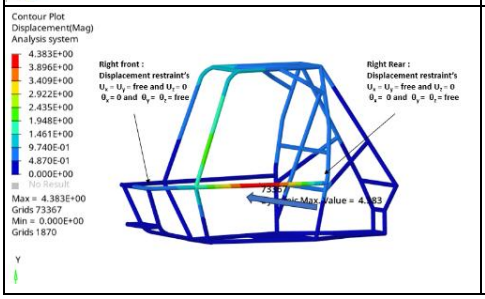
Bump Test



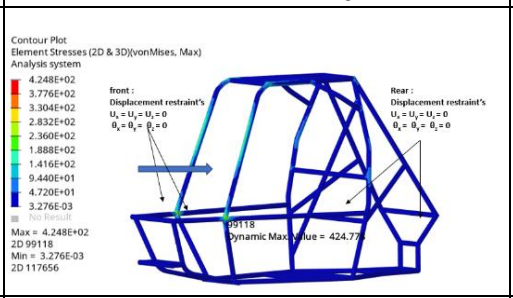


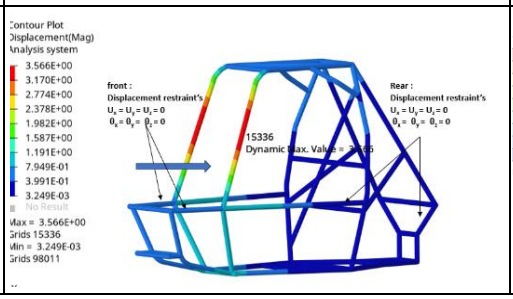
Side Impact



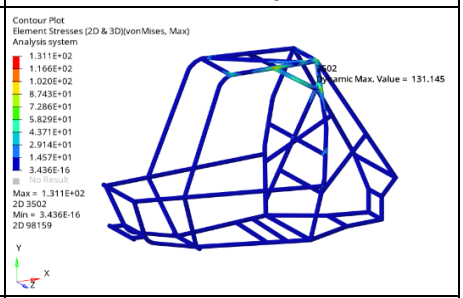


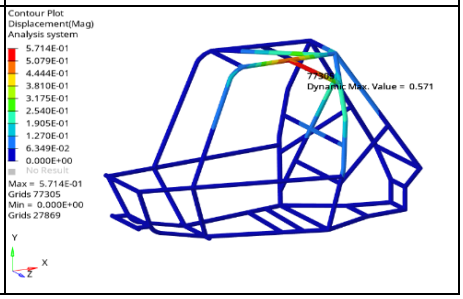
Pitch Over Analysis





Gusset Impact

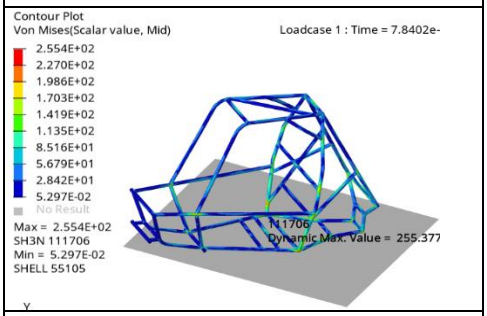
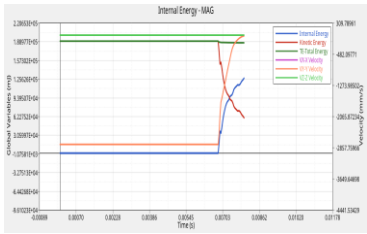


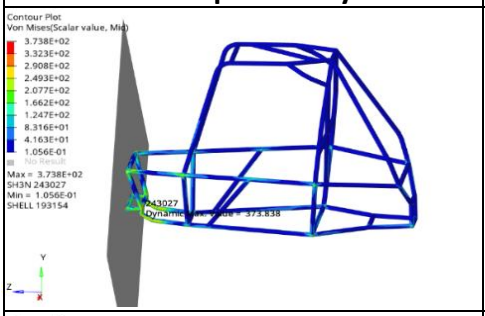
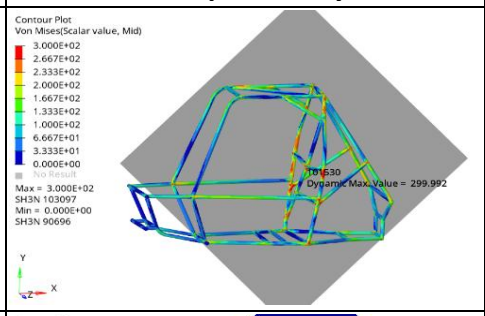
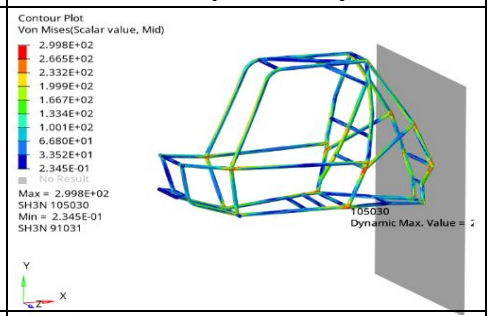
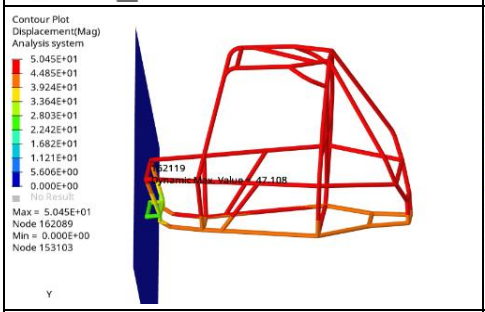
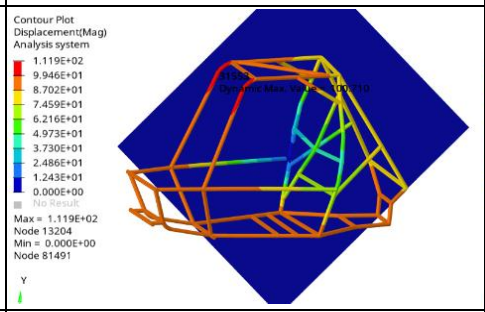
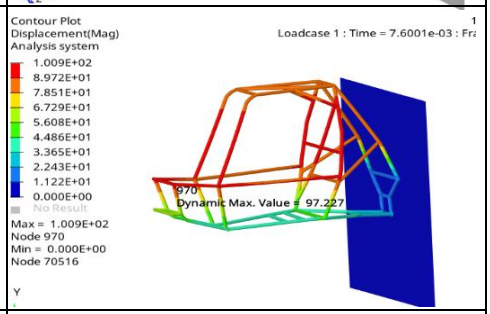


Impact	Front	Side	Torsional	Roll Over	Bump	Gusset
Load	12.24 KN	9.16 KN	83.80 KN	3.97 KN	7.34 KN	4.87 KN
Stress	214.7Mpa	414 Mpa	356.94 Mpa	454.77 Mpa	449.29 Mpa	131.14 Mpa
Deform.	2.49 mm	4.83 mm	6.18 mm	3.56 mm	8.68 mm	2.57 mm

ROLLCAGE ANALYSIS

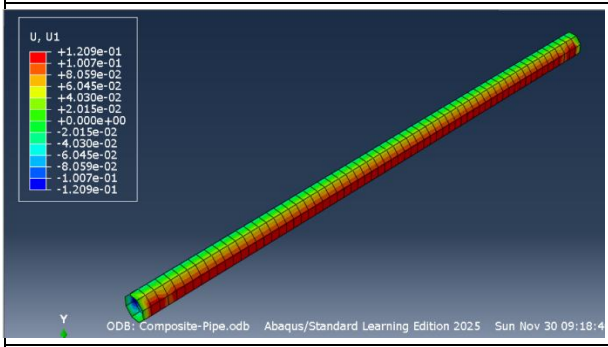
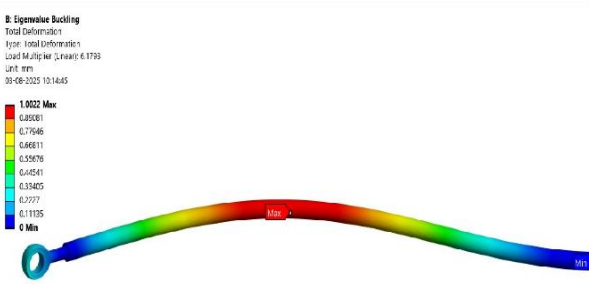
Dynamic analysis evaluates how the roll cage responds to **real crash conditions** where loads change rapidly. **RADIOSS** captures these transient effects to identify **peak stresses, energy absorption, and critical failure zones**, ensuring the roll cage can withstand severe impact condition and protect the driver.

Drop Test	Methodology for Dynamic	Graph Plotting
	1.T_{stop} chosen based on partial member recovery and stabilized kinetic/internal energy 2.The material model chosen was Johnson Cook material model with infinite rigid wall. 3.Type-7 contact is used for reliable handling of self-impact, buckling, and High velocity interactions. 4.Key interface parameters: Igap, Gapmin, INACTI = 6, N = 5, and Istf for interface stiffness.	 Internal Energy vs Time vs Kinetic Energy
Max Stress – 255 Mpa		

Front Impact Analysis	Side Impact Analysis	Rear Impact Analysis
		
		
Max Stress – 378 Mpa	**Max Stress – 300 Mpa**	**Max Stress – 299 Mpa**
Max Displacement – 5.04 mm	**Max Displacement – 1.11 mm**	**Max Displacement - 1.00 mm**

Composite Analysis - Extra Analysis (Non Linear Material)

ABAQUS was used for this analysis due to its robust nonlinear & composite-material capabilities, enabling prediction of deformation and structural response under loading.

Tie rod – Carbon Fiber	Tie rod – ASTM A36
	
Force – Compressive force of 1200 N	
Max Displacement – **1.20 mm**	Max Displacement **– 1.022 mm**

Conclusion - The carbon-fiber tie-rod shows slightly higher displacement than the ASTM A36 tie-rod but offers significant weight reduction while maintaining acceptable stiffness. Both designs remain within safe deformation limits for the applied load case.

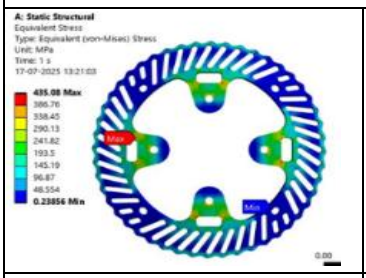
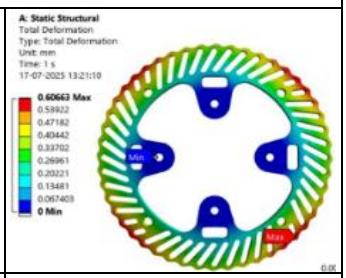
STATIC ANALYSIS – MATERIAL MODEL IS LINEAR

Static structural analysis is used to evaluate **stress**, **deformation & factor of safety** in a component under static loading Conditions.

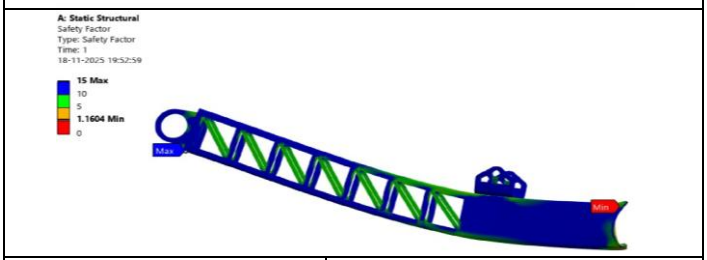
Methodology

1.CAD Part was made in Solidworks.
2.Imported in ANSYS by .iges
3. Geometry Cleanup was done
4.Mesh was performed (2mm Mesh Size)
5.Boundary Conditions Applied
6.Results (stress, displacement...etc.)

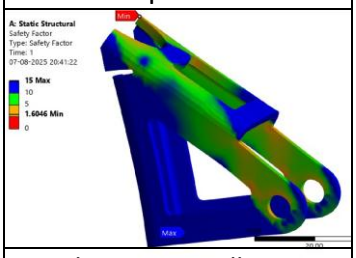
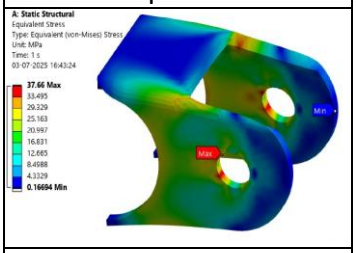
1. Brake Rotor - SS 410

Boundary Condition - While Braking	
	
Fixed Support	Applied at Bolt holes
Force & Moment – 2822.4 N , 359 Nm	Applied at Rotor Faces
Stress – 435.06Mpa	**Deformation – 0.6 mm**

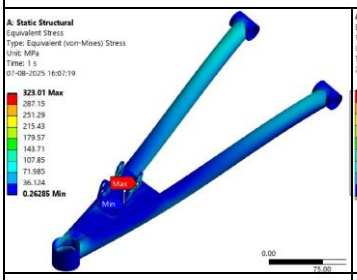
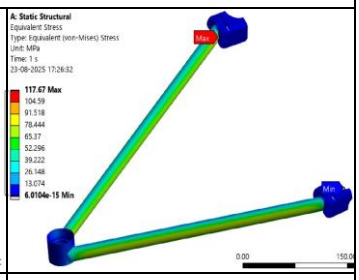
2. Trailing Arm – ASTM A36

Boundary Condition – One Wheel Landing			
			
Fixed Support		Mounting Points	
Force – 1500N		At Damper Mount	
Stress	**215.45MPa**	**Deformation**	**0.15mm**
FOS – 1.16			

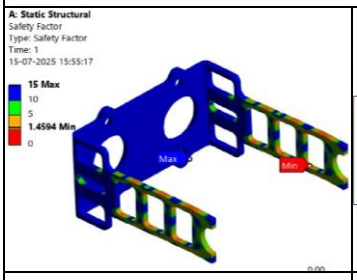
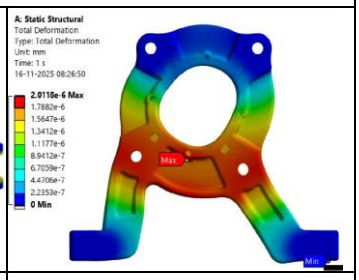
3. Front Upper & Rear Upper Damper Mount ASTM A36

Boundary Condition – Peak Bump Case	Boundary Condition – Peak Bump Case
	
Fixed Support–Roll Cage	Fixed Support–Roll Cage
Force – 2000N applied at MC bolt holes	Force – 2000N applied at MC bolt holes
FOS – 1.60	**FOS – 1.45**

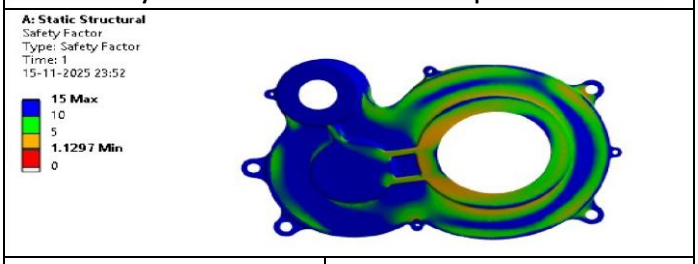
4. Upper & Lower Arms – AISI 4130

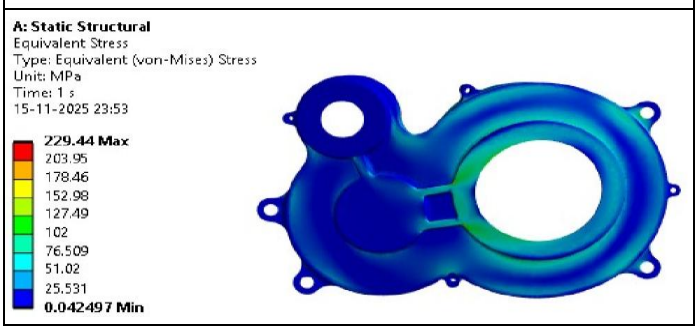
Boundary Condition – Single Wheel Bump	
	
Fixed Support – at mounting location	Fixed Support – at mounting location
Force – 4500N at Upper Ball Joint	Force – 2200N at Lower Ball Joint
Stress – 323.01 Mpa	**Stress – 117.67 Mpa**

5. MC Mount (MS) & Motor Mount (Al 7075)

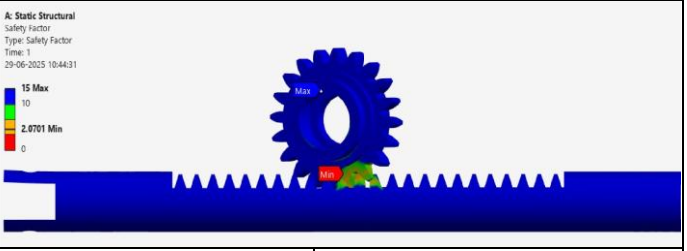
Boundary Condition – Max Loadings	
	
Fixed Support – Front Nose Tube	Fixed Support – at mounting location
Force – 2000N at MC Bore	Moment – 62 Nm at motor bolting
FOS – 1.45	**FOS – 1.62**

6. Gearbox Casing – Al 7075 T6

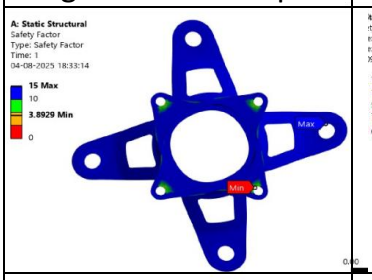
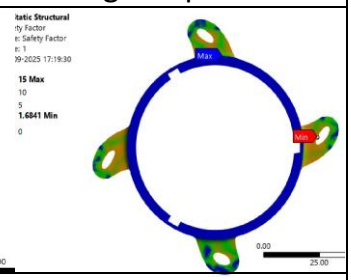
Boundary Condition – Lateral Impact			
			
Fixed Support		Applied at bolts	
Force – 2500N		At Spool OD	
Stress	**229.4 Mpa**	**Max Def.**	**0.595 mm**
FOS - 1.12			



7. Rack and Pinion – EN 19

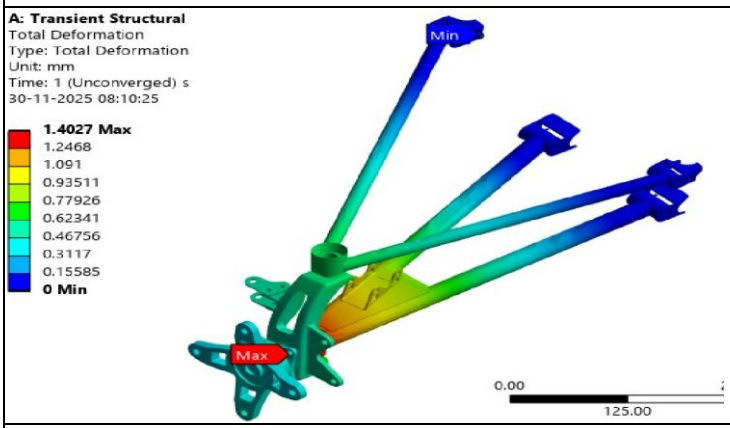
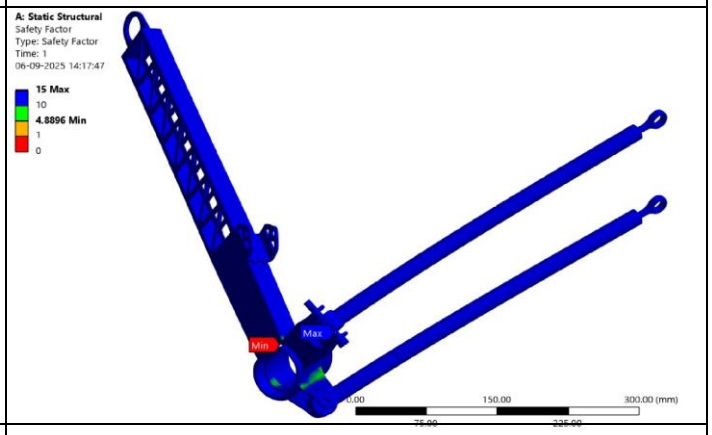
Boundary Condition – Lateral Impact			
			
Fixed Support		Applied at clevis bolts	
Moment – 50 Nm		At Pinion inner face	
Stress	**315.6MPa**	**Max Def.**	**0.1 mm**
FOS – 2.07			

8. Wheel Hub (Al 7075) & Brake Hub(AL 6065)

Boundary Condition - Single-Wheel Bump	Boundary Condition – Braking Torque
	
Fixed Support – At Bearing's Face	Fixed Support - At Inner Face
Force – 5000N	Moment – 319 Nm
FOS – 3.89	**FOS – 1.68**

Conclusion - All components were analysed under maximum static loading, and the results confirmed that the design meets the required strength criteria with an optimum factor of safety.

DYNAMIC ANALYSIS

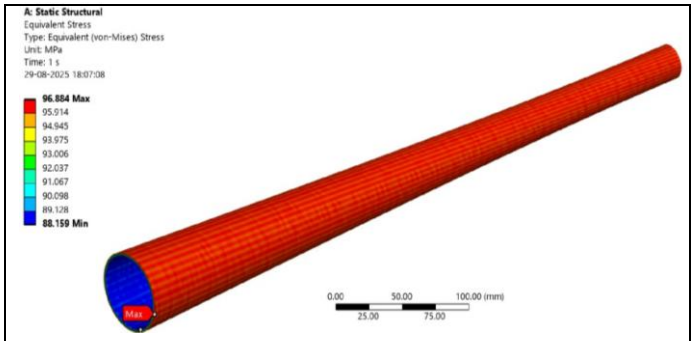
Front Assembly	Rear Assembly
	
Max Deformation – 1.40 mm at varying loads	**FOS – 4.88**

Conclusion - Dynamic analysis of the front and rear suspension assemblies verified that the system performs safely under varying road loads and dynamic inputs, maintaining structural integrity and achieving an optimum factor of safety or minimum deformation throughout the operating conditions.

Torsional & Bending Analysis

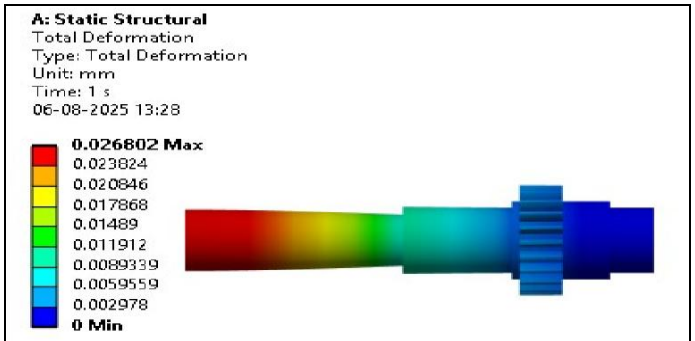
Torsional and bending analyses were performed to evaluate the component's **resistance to twisting, vertical and lateral loads.** These studies ensured adequate rigidity, controlled deflection, and reliable structural

1.Steering Column (Torsional)



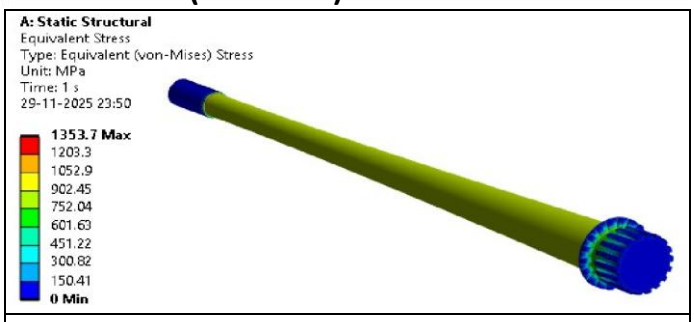
CALCULATIONS		
	Analytical	FEA
Max Shear Stress	τmax = (T × r) / J = **55.6 MPa**	**96.88 MPa**
Angle of Twist	θ = (T × L) / (G × J) **=0.029 rad**	-

2. Input Shaft(Torsional)



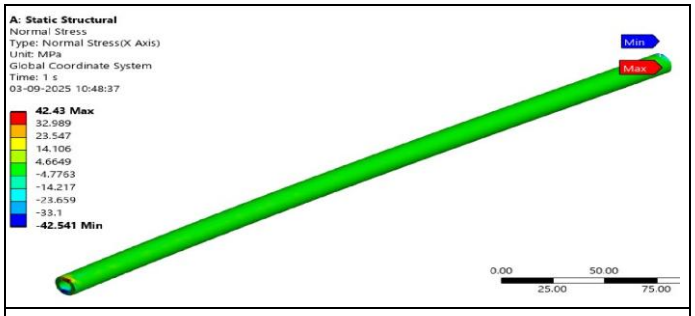
CALCULATIONS		
	Analytical	FEA
Max Shear Stress	τmax = (T × r) / J = **97.27 MPa**	**103.3 MPa**
Angle of Twist	θ = (T × L) / (G × J) **= 0.071 rad**	-

3. Half Shaft (Torsional) – EN 19



CALCULATIONS		
	Analytical	FEA
Max Shear Stress	τmax = (T × r) / J = **1124.72 Mpa**	**1353.7 MPa**

4. Camber Link(Bending) - AISI 4130

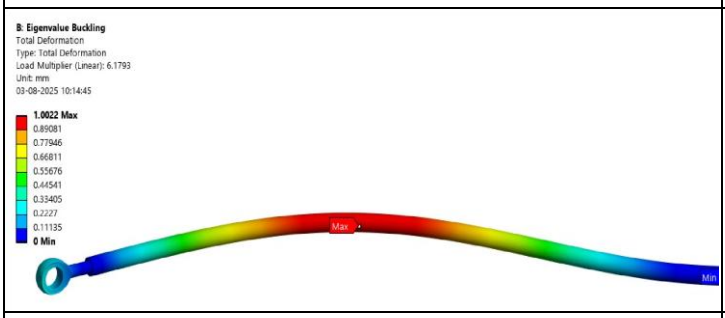
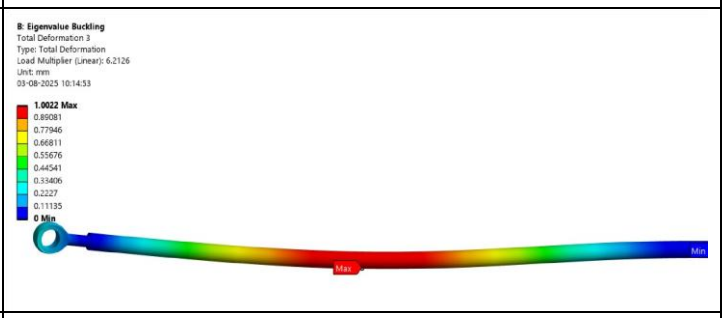


CALCULATIONS		
	Analytical	FEA
Max Shear Stress	τmax = (T × r) / J **= 33.4 MPa**	**42.43 MPa**

Conclusion - Torsional and bending analysis confirmed that all components safely resist applied twisting and bending loads, maintaining required stiffness and structural integrity. These results helped validate the design and ensured <u>each part meets performance and safety requirements.</u>

EIGEN BUCKLING – EXTRA ANALYSIS

Eigen buckling analysis predicts the load at which a component becomes **unstable under compression**.

Boundary Condition – Lateral / Cornering Impact	Modes Conducted - 6
	
Max Deformation – 1.022 mm	**Avg. Deformation – 0.44 mm**

Conclusion - The lateral load analysis <u>confirmed pulsating safe stress levels and controlled deformation on Camber Links</u>, while the Eigen-buckling study across multiple modes verified adequate stability, ensuring the component's structural reliability.

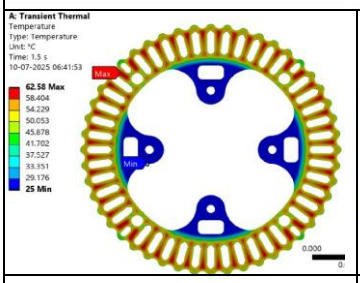
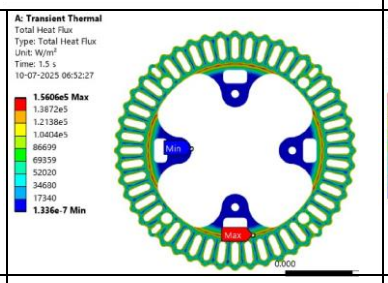
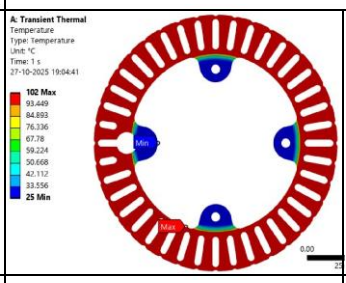
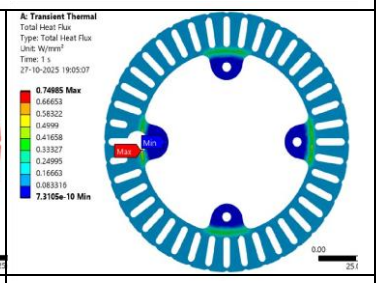
THERMAL ANALYSIS

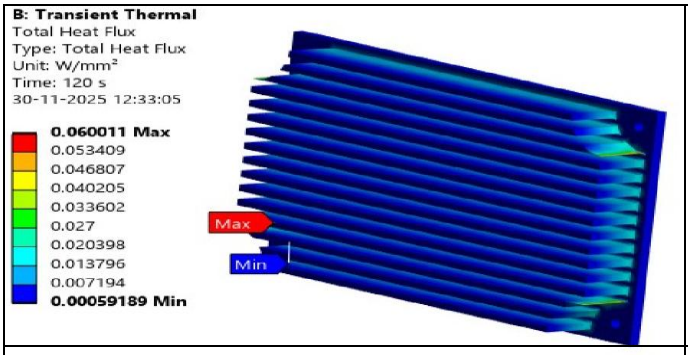
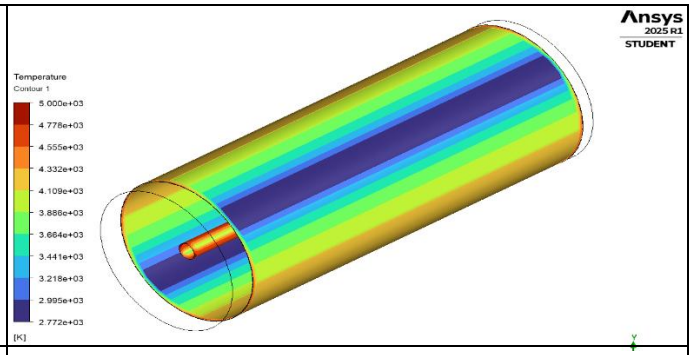
Thermal analysis evaluates temperature distribution and heat flow within a component under steady or transient thermal loads. It's performed to understand how heat affects material behaviour and performance.

Methodology Steps for Thermal Analysis

1. Import/Create Geometry ⟶ CAD model of the part.
2. Define Material Properties ⟶ Thermal conductivity, specific heat, density, etc.
3. Meshing ⟶ Discretize the geometry into elements for analysis.
4. Apply Boundary Condition ⟶ Temperature/Heat flux / convection / radiation/Heat generation
5. Solve ⟶ Run the thermal solver to calculate temperature distribution.
6. Post-Processing ⟶ Plot the Temperature contours, heat flux, gradients.

1. Brake Rotor

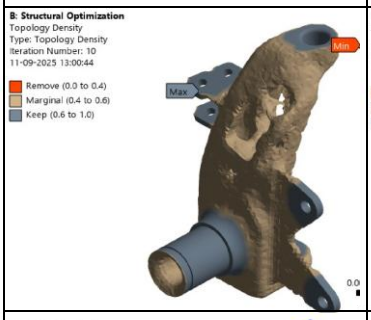
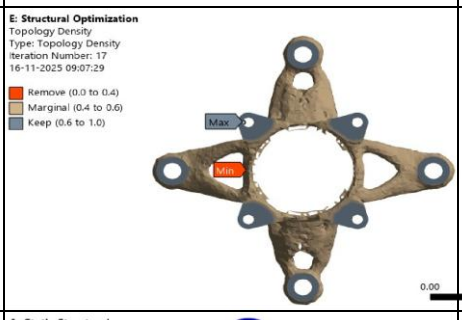
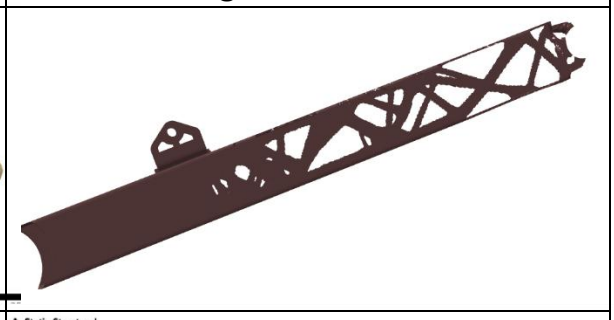
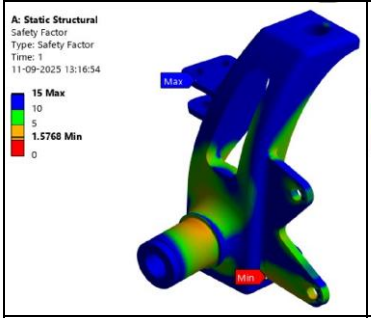
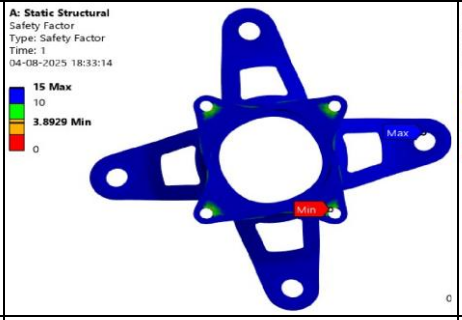
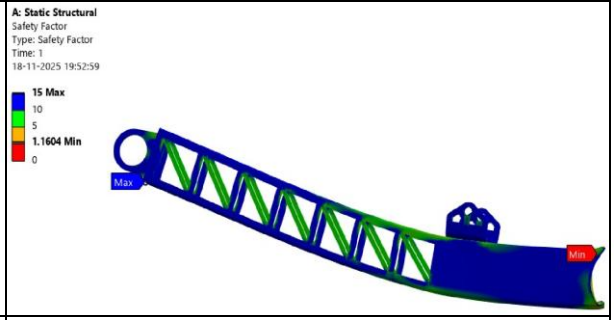
Front Rotor – SS410		Rear Rotor – SS410	
			
Temp. Rise – 62.58°C	**Heat Flux – 1.56e5 W/mm²**	**Temp. Rise – 102°C**	**Heat Flux - 0.759 W/mm²**

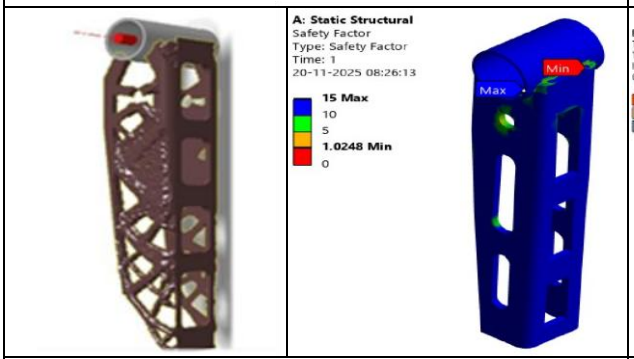
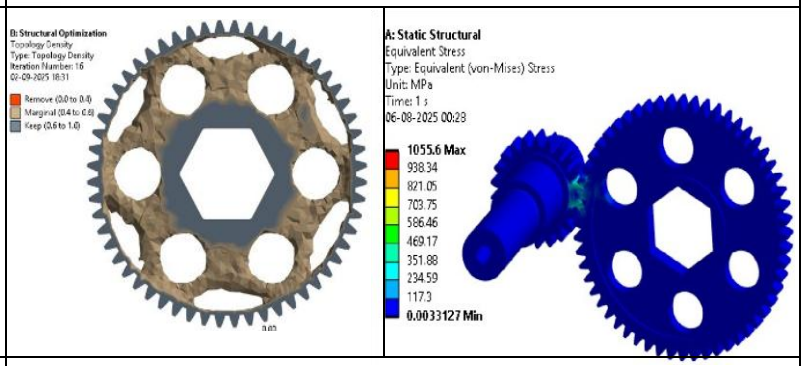
3. Heat Sink – Al 6065 T6	4. Battery Cell – BAK NMC 21700
	
Heat Flux – 0.06011 W/mm²	**Temperature generated – 50 °C**

Conclusion - The thermal analysis confirmed that <u>all components remain within safe temperature limits under operating heat loads.</u> Brake rotors showed the highest temperature rise but stayed structurally stable, while battery cell & Heat Sink exhibited uniform heat distribution ensuring safe performance.

TOPOLOGY ANALYSIS

Topology optimization identifies optimal material distribution along load paths, helping **reduce weight** while **maintaining required strength** and stiffness.

Knuckle – Al 7075 T6	Wheel Hub – AL 7075 T6	Trailing Arm – ASTM A36
		
		
Weight Reduced – 175 g	Weight Reduced – 220 g	Weight Reduced – 420 g
FOS after Topology - 1.5	**FOS after Topology – 3.89**	**FOS after Topology -1.16**

Brake Pedal – ASTM A36	Gear – EN 36
	
Weight Reduced – 175 g	Weight Reduced – 164.17
FOS – 1.02	**FOS – 1.51**

Conclusion - Topology optimization achieved significant mass reduction across components, with <u>weight savings of **11.25 %.**</u> All redesigned parts retained safe FOS values, confirming structural reliability after material removal.

MODAL ANALYSIS

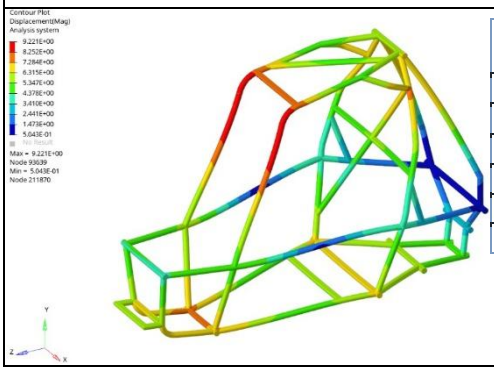
Modal analysis determines how **a structure vibrates by extracting its natural frequencies**. This ensures the component remains safe from resonance under real operating conditions.

Methodology Steps for Modal Analysis

1. Import/Create Geometry ⟶ CAD model of the part/assembly.
2. Define Material Properties ⟶ Young's modulus, density, Poisson's ratio, damping (if applicable).
3. Meshing ⟶ Discretize the geometry into finite elements suitable for dynamic analysis.
4. Apply Boundary Conditions ⟶ Realistic constraints such as fixed supports, joints, or mounting interfaces.
5. Solve ⟶ Run the modal solver to extract natural frequencies and corresponding mode shapes.
6. Post-Processing ⟶ Plot mode shapes, identify critical vibration modes, and compare with excitation frequencies.

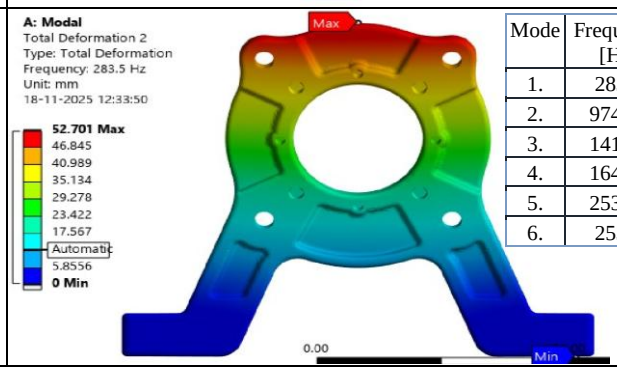
NOTE - The DAQ system captured a vibration frequency of <u>100 Hz</u> under real-world operating conditions.

Chassis – AISI 4130



Mode	Frequency [Hz]
1.	23.20
2.	21.47
3.	14.93
4.	12.22
5.	7.77
6.	4.32.

Motor Mount – AL 7075 T6



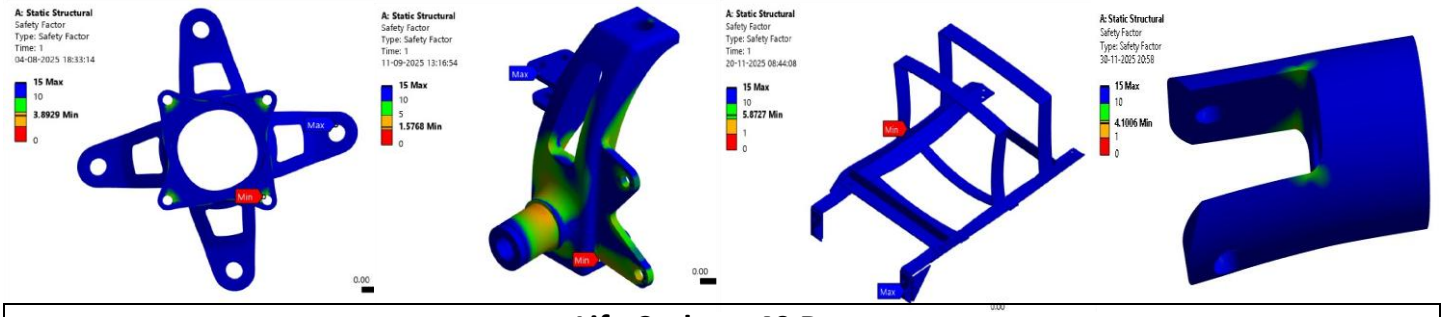
Mode	Frequency [Hz]
1.	283.5
2.	974.64
3.	1414.3
4.	1641.1
5.	2530.7
6.	2533.

Conclusion - Modal results confirm safe frequency margins and vibration stability. <u>Both the Components are not falling near the Resonant Frequency, considering real life error **± 10%.**</u>

FATIGUE ANALYSIS

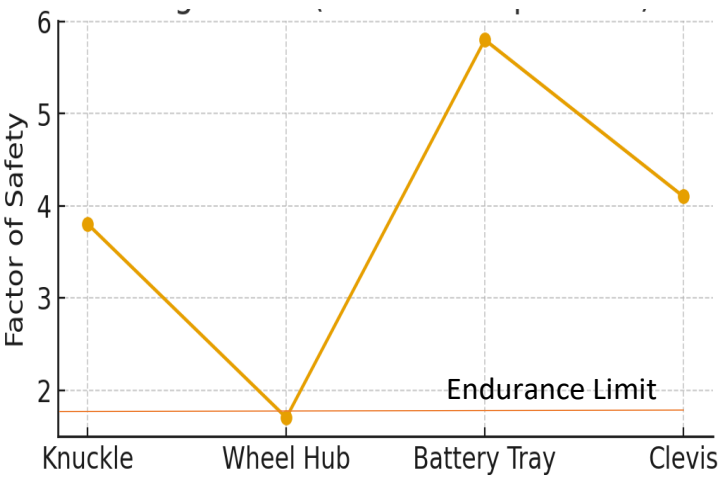
Fatigue analysis predicts the life of a component under **repeated or cyclic loading** by estimating when **cracks or failures** might initiate. It's performed to ensure durability and prevent unexpected failure over time.

Wheel Hub	Knuckle	Battery Tray	Clevis



Life Cycles – 40 Days				
FOS	**3.8929**	**1.5768**	**5.8727**	**4.1006**

Graph



Endurance Limit – 1.5

An endurance limit of 1.5 was selected as a conservative safety margin for reliable performance under cyclic loading

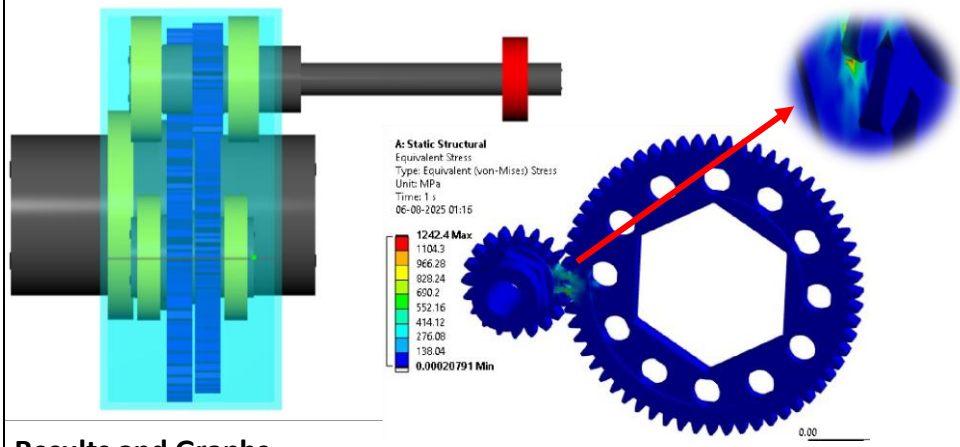
Conclusion –

The fatigue analysis confirmed the structural reliability of the wheel hub, knuckle, clevis, and battery tray under expected cyclic loading. All components achieved factors of <u>safety above the endurance limits</u>, demonstrating compliance with durability criteria and ensuring <u>no risk of fatigue</u> -related failure <u>during operation.</u>

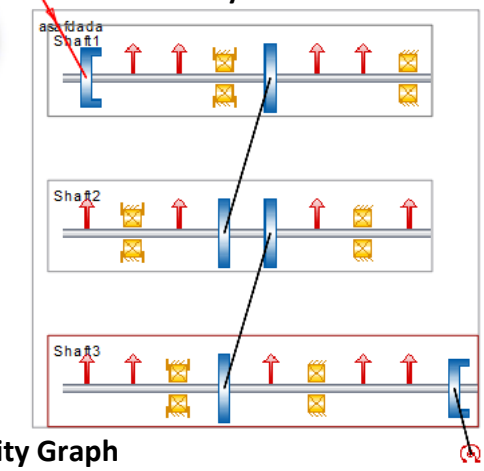
POWERTRAIN SIMULATION & ANALYSIS

Powertrain Efficiency – KISSsoft & Ansys

KISSsoft is a professional transmission-design and simulation software used to evaluate gear performance, power losses and overall system efficiency. We used KISSsoft to calculate the **powertrain efficiency** of our BAJA drivetrain, considering gear meshing losses, bearing losses, churning losses, and seal losses.



Layout

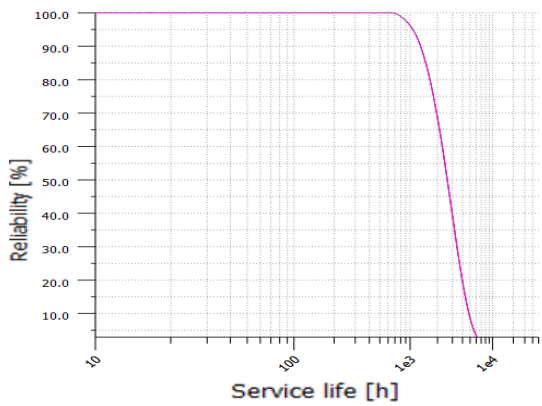


Results and Graphs

Efficiency and Heat Results -

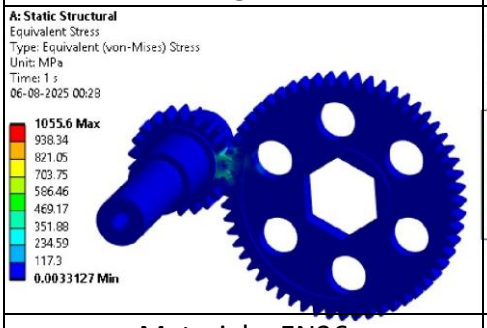
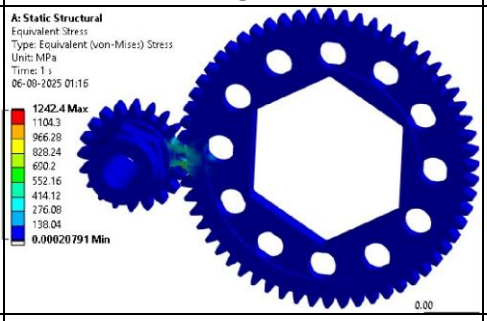
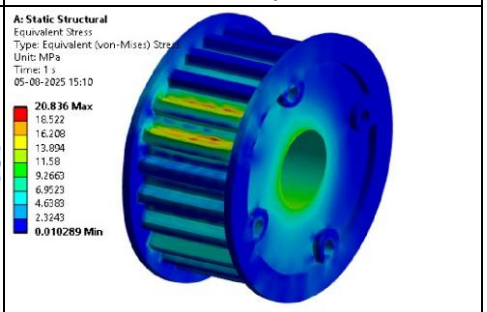
	A	B	C	D	E
1	.				
2	Requested Result	Heat Transfer		Units	W
3	Calculation Method	ISO TR 14179-2		Contact Analysis	No
4					
5	Housing	Settings	Ventilation	Lubrication	Oil Cooler
6					
7	SPECIFIC FUNCTIONS				
8	Reset		Consistent		Report
9					
10	RESULTS SUMMARY				
11	Power Input [W]	23561.9		Efficiency (total) [%]	96.04
12	Power Output [W]	22629.1		Efficiency (gear mesh) [%]	97.25
13	Power Losses [W]	932.8		No Req. Result	0
14					
15	LOSSES AND DISSIPATIONS				
16	Heat Generated	Correction Factors		Heat Dissipated	Correction Factors
17	Gear Churning Losses PVZ0 [W]	170.6		Housing [W]	52.8
18	Gear Meshing Losses PVZ [W]	649.1		Foundations [W]	0
19	Bearing Losses PVL+PVL0 [W]	113.1		Input/Output Shafts [W]	59.1
20	Seal Losses PVD (+rest PVD0) [W]	0		Oil Cooler [W]	0
21	Total Power Losses [W]	932.8		Total Dissipation [W]	111.9

Reliability Graph



Results -

1. Overall Efficiency – **96.04 %**
2. A gear efficiency of **97.75%** means that only 2.25% of power is lost in tooth contact, mostly as heat due to friction.
3. Reliability v/s Service life Graph

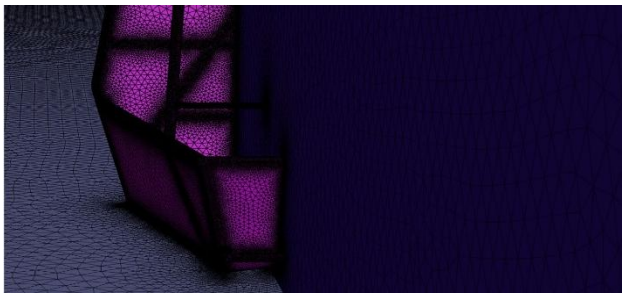
1st Stage Gear	2nd Stage Gear	Pulley
		
Material – EN36	Material – EN36	Material – Al 7 series
Max Stress – 1055.6 Mpa	**Max Stress – 1242.6 Mpa**	**Max Stress – 20.836 Mpa**

Conclusion – The powertrain analysis confirmed high overall efficiency through KiSSsoft-based gear evaluation, with minimal losses from meshing and bearings. ANSYS FEA validated that all gears and pulleys remain structurally safe under operating loads, and the reliability assessment demonstrated long service life, ensuring a robust and efficient drivetrain design.

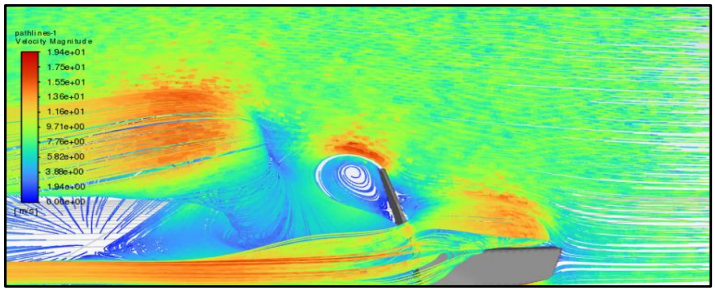
CFD ANALYSIS

1. Methodology steps for Chassis CFD -

1. The chassis CAD was created in SolidWorks and imported into ANSYS Fluent.
2. A half-model with a symmetry plane was used for faster computation.
3. Inlet, outlet, and symmetry boundary conditions were applied.
4. The standard k-ε turbulence model was used
5. Mesh refinement with prism layers was generated.
6. Pressure distribution and drag force were obtained.



Inlet	*16.67 m/s*
Turbulence model	*SST k–ω*
First cell thickness	*y+=1 → 2.11e−5 m; y+=30 → 6.32e−4 m*
y+ target (k–ε)	30 – 300

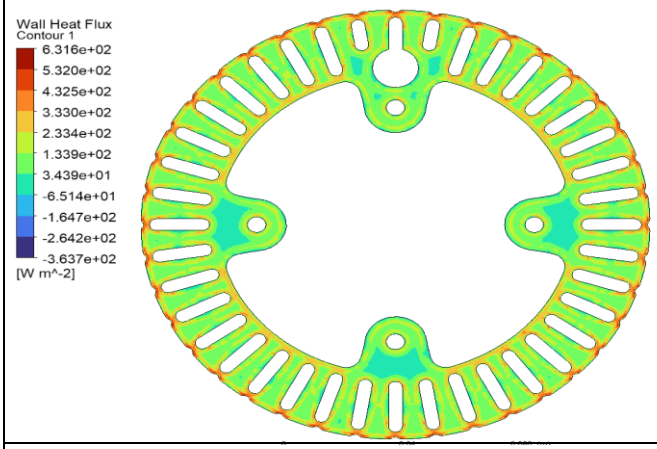


Conclusion - The pressure plot shows high pressure at the front and a low-pressure wake behind the chassis, indicating proper flow separation. The drag graph shows a peak drag of about 90–92 N, confirming the expected aerodynamic load

2. Methodology steps for Rotor's -

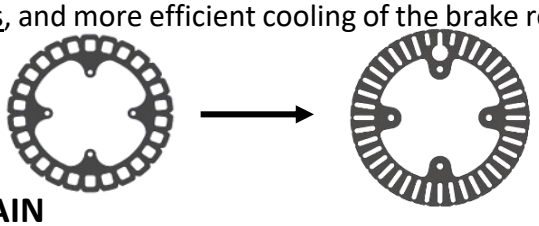
1. Rotor geometry was prepared in **SolidWorks** and imported into ANSYS Fluent**.**
2. The K- omega SST was used for accurate heat-transfer prediction.
3. Inlet, outlet, and wall heat-flux conditions were applied to represent braking.
4. The simulation was solved to capture rotor cooling behaviour.
5. CFD-Post was used to extract the convective heat-transfer coefficient.

Solver – Ansys Fluent	MODEL – K OMEGA SST
Boundary Condition – While Braking at **1.33** sec	
Inlet Velocity - 16.67 m/s	Ambient Temp. – 27 °C
Max Temperature – 73 °C	



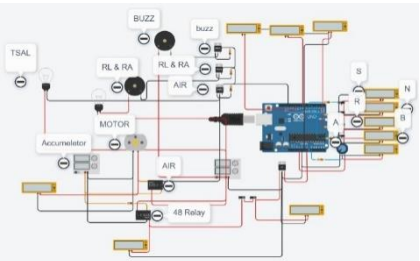
Wall Heat Flux – 6.31e² Wm²

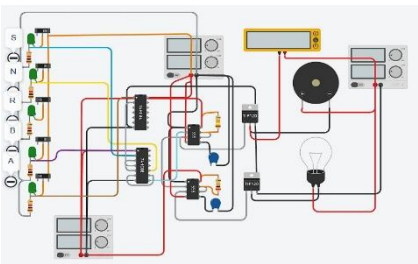
Conclusion - Through CFD, we evaluated the rotor's thermal performance during braking and modified the slot design for better airflow circulation. This resulted in improved heat removal, reduced hot spots, and more efficient cooling of the brake rotor.



E POWERTRAIN

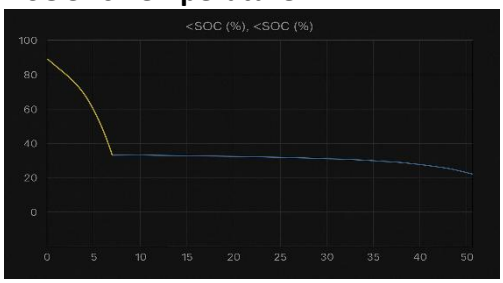
Microcontroller GLV Circuit vs Logic Gate GLV Circuit





Conclusion - TinkerCAD validation showed that the logic-gate circuit became complex and difficult to modify as more safety conditions were added. The microcontroller circuit offered a cleaner design with fewer components, easier updates through programming, and higher reliability. Hence, the microcontroller-based GLV circuit was chosen for its simplicity, flexibility, and better overall control.

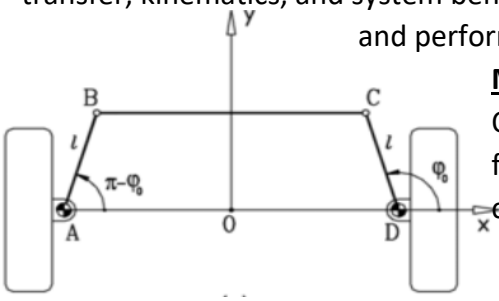
SOC vs Temperature - MATLAB



Conclusion - The SOC graph shows an initial steep drop followed by a slow, steady decline, indicating that the battery experiences a brief high load before settling into stable discharge over time.

Multi Body Dynamics

Multibody Dynamics (MBD) is a simulation method used to study the motion and interaction of interconnected rigid or flexible bodies under loads and constraints. It helps engineers understand force transfer, kinematics, and system behaviour without physical testing, providing insights into stresses, fatigue, and performance for design validation and optimization.



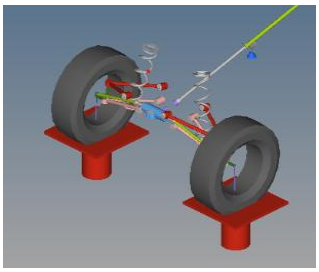
Mechanism – Four Bar Linkage

Our vehicle uses a double-wishbone suspension, which functions as a four-bar Linkage mechanism to precisely control wheel motion and camber characteristics.

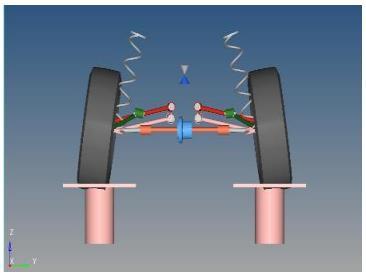
Full Vehicle Simulation in Motion View -

MotionView enables detailed multibody simulation of the suspension, helping evaluate wheel motion, load transfer, and kinematic characteristics without physical prototypes.

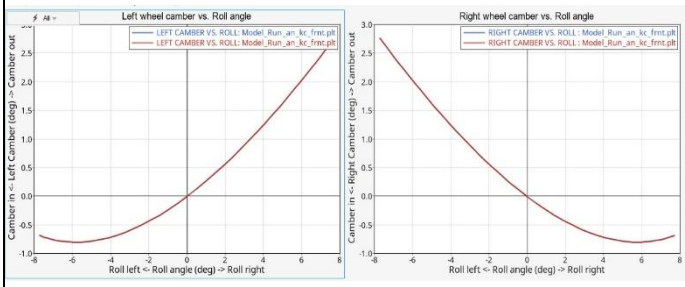
Front suspension	*Camber: 3°*	*Toe angle: 2°*
	Caster: 4 °	*Kpi: 9°*
Rear suspension	*Roll center: 292 mm*	*Toe angle:4°*
	Track Change: 87mm	*Camber : 6°*
Front & Rear Wheel Travel : 10.66 in		



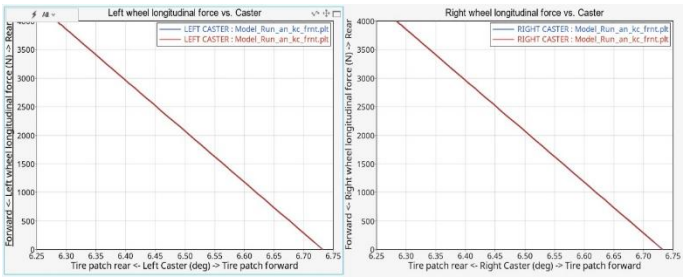
Front simulation



Rear simulation



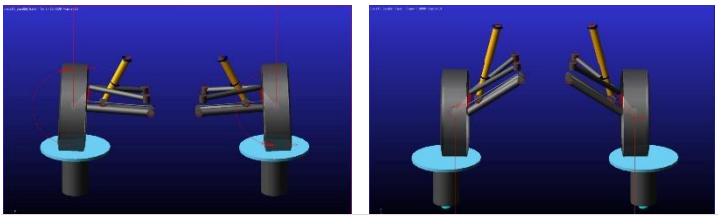
Wheel Camber Vs Roll angle



Wheel longitudinal force Vs Caster

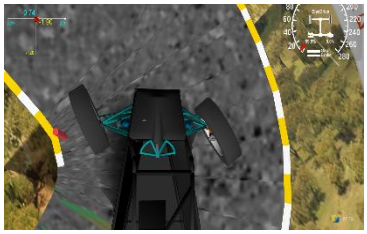
Result - The camber-roll curves show a smooth and predictable camber gain on both wheels, indicating stable lateral geometry during cornering. The caster-force plots exhibit a linear variation in longitudinal forces, confirming correct steering axis behaviour and symmetric response between left and right wheels.

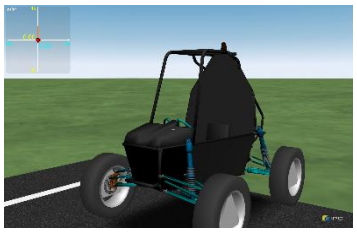
Front Geometry Simulation in MSC ADAMS

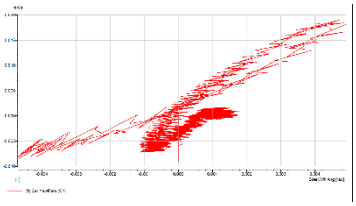


Conclusion - The front LCA force reduces steadily from approximately 4000 N to near zero mid-travel and then increases gradually, indicating a consistent and realistic suspension load response throughout the wheel-travel cycle.

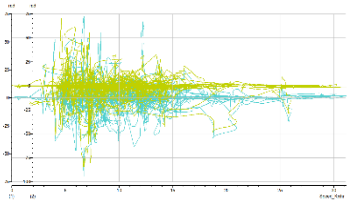
IPG CARMAKER



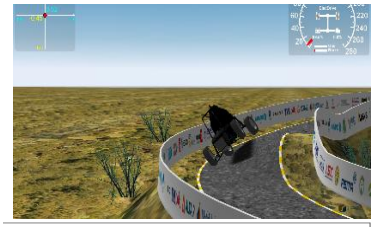


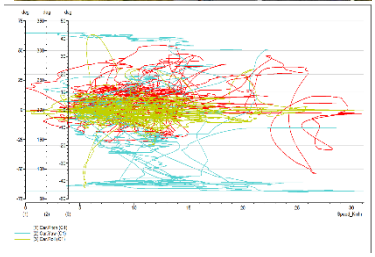


Steer Wheel vs Yaw Rate

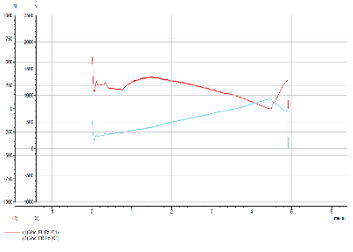


Speed vs Front Slip Angle





Roll, Pitch, Yaw vs Speed



Front L & R Load vs Time

OVERALL CONCLUSION

This year's CAE work covered a wide range of analyses across multiple vehicle subsystems, including wheel hubs, uprights, brackets, powertrain mounts, and chassis interfaces. The study included structural, modal, torsional – bending, fatigue, Eigen-buckling, and multibody dynamics evaluations to ensure all components met performance and safety requirements. Along with the standard workflow, we implemented advanced methods such as topology optimization, improved mesh convergence checks, and refined kinematic validation. These additions increased accuracy, enhanced design understanding, and supported more effective decision-making during component development.

Component	Types of Analysis	Acceptance Criteria	Statement	Conclusion
Trailing Arm	Static	Max Stress < Material YTS	215.45 < 410 MPa	Design is Safe
Heat Sink	Thermal	Max Temp. < Cut off Temp	64 < 85 °C	system stays active and does not shut down
Wheel Hub	Topology	FOS ≥ Endurance limit	3.82 ≥ 1.5	Wheel Hub is Safe
Motor mount	Modal	Natural frequencies must not match or fall close to excitation frequencies.	Natural frequencies are above excitation range, preventing resonance.	Frequencies > excitation → no resonance.
Gearbox	Power Loss & Efficiency Analysis	Efficiency must remain above required operating threshold	Software evaluated losses from gears, bearings, and lubrication	Achieved acceptable efficiency for required performance

Brakes

→ CFD-based HTC analysis of last year's heavier, Low slot rotor showed poor cooling, which helped us redesign the new rotor with more slots while maintaining effective heat dissipation.

Suspension & Steering

→ Suspension analysis, supported by DAQ-measured forces, helped identify critical and non-critical load regions enabling effective topology and structural optimization to reduce weight while maintaining strength and kinematic performance.

E Powertrain

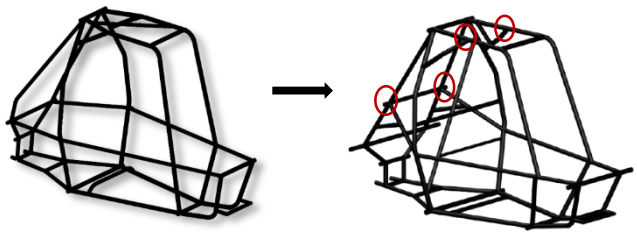
→ Thermal analysis helped optimize the heatsink design by improving airflow contact and reducing temperature rise. Additionally, virtual validation on TinkerCAD ensured correct circuit behaviour and power regulation.

Transmission

→ We evaluated gear efficiency and losses using KISSsoft and verified structural safety through ANSYS FEA. These analyses helped us refine gear geometry, improve mesh efficiency, and ensure all powertrain components operate reliably under load.

Chassis

→ With the HyperMesh, we identified the tubes with low load contribution, enabling reduction of bends while keeping structural rigidity intact.



2025 Season	2026 Season
16 Bends	12 Bends

Software's Used

Ansys	Structural,Dynamic,Thermal,Modal,CFD & Fatigue Analysis
Altair Inspire	Topology & Static Analysis
Altair Hyperworks	Rollcage Analysis
Solidworks	Kinematic Visualisation
Kissoft	Powertrain Efficiency Calculations
MATLAB	Battery Simulations
TinkerCAD	Virtual Validation of Circuit
MSC Adams	Suspension Kinematics & Multibody Dynamics
IPG Carmaker	Full Vehicle Dynamics Simulation
ABAQUS	Composite Analysis